

GUANNAN LYU

BUSINESS ARCHITECTURE SPECIALIST

Location: Toronto, Canada

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accenture technology
Intelligent Engineering Services

PROFESSIONAL BACKGROUND

Guannan Lyu is a Business Architecture Specialist and full-stack engineer with strong experience in backend development, cloud-native modernization, and high-impact mobile solutions. He has led critical features, modernized legacy systems, and delivered cloud-native architectures on GCP and Anthos. Known for translating complex requirements, mentoring developers, and consistently delivering reliable, maintainable systems recognized by leaderships.

FUNCTIONAL EXPERIENCE

Backend Engineering. Full-Stack Development. Cloud-Native Architecture. Microservices Design. Mobile Platform Development. System Modernization & Migration. Application Performance Optimization. Data Processing & Integration.

TECHNICAL EXPERIENCE

Java, Spring Boot, Spring Security, Typescript, Python, C#, Node.js, GCP, Firebase, Azure, Anthos, Kubernetes, Microservices, Apache Kafka, Angular, React, SQL Server, Hibernate, MyBatis, PostgreSQL, .NET Core, Git, Apache JMeter

INDUSTRY EXPERIENCE

- Transportation Logistics
- Aerospace
- Technology & Digital Platforms

CERTIFICATIONS

- Microsoft Azure Fundamentals (AZ-900)
- Oracle Certified Associate, Java SE 8 Programmer

SELECTED RELEVANT EXPERIENCE

Senior Full Stack Developer / Tech Lead – CN Rail (Backend & Mobile Platforms)

- Led end-to-end delivery of multiple mission-critical features on cross-platform mobile apps and backend microservices. Designed and implemented cloud-native solutions using GCP and Firebase, collaborating with client teams to resolve technical constraints and align architectural decisions. Guided developers and QA analysts on implementation approaches, testing processes, and best practices for quality and performance.
- Improved application stability in low-connectivity environments through offline-first patterns, background data synchronization, and connection-quality monitoring. Led the migration of an internal mobile app to a modern GCP-based architecture, including authentication integration, dependency upgrades, and platform compatibility trouble-shooting.
- Served as the primary technical owner for migrating critical backend microservices from legacy infrastructure to a cloud-native Spring Boot architecture. Redesigned parsing, search, and data-handling logic, delivering significant performance boost and reducing long-term maintenance complexity. Defined service architecture, integration patterns, data models, and deployment structures for cloud-based environments.
- Consistently improved code quality, runtime performance, and stability through refactoring, pull request code reviews, and data structure optimization. Performed load testing using automation tools.
- Recognized by client leadership for strong ownership, timely delivery, and high-impact technical contributions. Provided mentorship and architectural guidance to new development team members.

Full Stack Developer - Airbus

- Improved the performance of a key internal application by optimizing DOM rendering and backend–frontend data flow, reducing load time by over 70%.
- Delivered enhancements across front & backend, optimizing API logic, refining UI components, and improving data-flow efficiency across the system.